Final Project – Group Work

Analyzing Cars Features to Predict their Prices

A Regression Based Approach Using UCI Data

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Abstract

This study explores key cars attributes that influence car prices using the *Automobile Dataset*. The dataset includes 205 car records with 26 variables, including categorial information (fuel type, body style) and numeric (engine size, horsepower, curb weight). After cleaning missing values and converting relevant fields into appropriate data types, we ran exploratory data analysis using correlation, scatterplots, and summary statistics to examine primary relationships, detect outliers and chose the model variables.

The multiple linear regression model that we build using both quantitative and qualitative predictors explains approximately 92.5% of the variance in price indicating a strong fit. Our model includes statistically significant variables such as engine size, curb weight, highway MPG, and rear engine location (strong positive correlation) and variables such as certain body styles and lower cylinder counts (strong negative correlation). Model diagnostics such as residual plots and Cook's distance were used to evaluate model fit and detect influential observations. Our model provides valuable insights into which car features most strongly affect pricing. The findings can support better data-driven decision-making for both consumers, dealers and manufacturers and on the other hand can support better design, marketing strategies, and informed purchasing decisions.

Introduction

The purpose of this report is to analyze the factors that influence automobile prices by using a real-world dataset provided by the UCI Machine Learning Repository. Knowing what kind of technical or categorial features contribute most to car pricing and understating the relationship can help manufacturers, dealers, and consumers make more informed decisions when selling or buying a car.

This regression analysis uses a combination of numerical and categorical variables (Dummy variable) to determine which car attributes, such as engine size, horsepower, fuel type, automobile body type etc. are the most important predictors of car costs. To build and evaluate our model, we use statistical methods such as exploratory data analysis, multiple linear regression modeling, and residual analysis.

Our report starts with describing the dataset and its data characteristics, and outlining its cross-sectional, observational nature. To better examine trends, spot any outliers in our model, and choose which variables to include, we used scatterplots, and summary statistics to show the data. Then, we built the model interpreted the coefficients and the overall usage of the model for predictions of the cars price. We also look closely at multicollinearity, heteroscedasticity, and data points that strongly affect the model. We close our report conclusions and recommendations based on our findings, limitations and suggestions for further analysis.

Data Characteristics

The dataset, titled "*Automobile Data Set"* is from UCI Machine Learning Repository public datasets. This dataset includes a wide range of attributes (26 variables) for a total of 205 records (205 car models). Each row represents a car model and includes both categorical and numerical features such as fuel type, car body, engine size, horsepower, and price. Our dataset is cross sectional because it is collected at one point in time for different cars rather than tracking changes over time and its observational (not experimental) because there is no manipulation or control over the variables as we only analyze existing data. The dataset was well-structured and required no heavy data cleaning. We dropped 8 rows with missing values retaining 197 complete observations.

We built scatter plots for all numeric variables to visualize relationships between predictors and price. These plots help understand linear trends or not, for example, engine size, curb weight, and horsepower show strong positive

relationship with price, while variables like city mpg show negative trends. We tried to also identify outliners and data skewness, especially in the price variable, which exhibits a right-skewed distribution with several high-end luxury vehicles pulling the mean upward. We also calculated correlations and decided to include in the model variables with high positive/negative correlation to price and with a strong linear relationship seen in the scatterplots (observations close to the linear regression lines), which will be listed and detailed below. Only plots and summary statistics most relevant to variable selection are included in this section, while additional exploratory visuals and model equation appear in the Appendix for reference together with a full list of variables and coding script.

Model and Interpretation

Before building the model, we analyzed the available variables by calculating their correlation with price and we also visualized the relationship with different scatterplots. We used these correlations to decide which variables to include in our model. Strong predictors like curb weight, engine size, and horsepower were included. Variables with weak correlations, like stroke and compression ratio, were excluded to keep the model efficient. We then constructed a multiple linear regression model, where price is the dependent (response) variable, and the predictors include both quantitative and categorical variables related to a vehicle design and performance. The purpose of the model is to identify which features significantly impact car prices and how much the effect of changes in these features on the price is.

Quantitative Variables Included

- **Wheel Base**: Distance between the front and rear axles
- **Length**: Overall length of the vehicle
- **Width**: Overall width of the vehicle
- **Curb Weight**: Weight of the vehicle without occupants or cargo
- **Engine Size**: Total volume of all engine cylinders
- **Bore**: Diameter of the engine cylinders
- **Horsepower**: Power output of the engine
- **City MPG**: Fuel efficiency in city driving conditions (miles per gallon)
- **Highway MPG**: Fuel efficiency in highway driving conditions (miles per gallon)
- **Peak RPM:** Engine revolutions per minute at peak horsepower

Categorical Variables and Dummy Variable Assignment

- In this section we converted all categorical (qualitative) variables into dummy variables, so they can be incorporated in the regression model:
- **Fuel Type:** 1 = gas, 0 = diesel
- **Aspiration:** 1 = turbo, 0 = standard
- **Number of doors:** 1 = two doors, 0 = four doors
- **Engine Location:** 1 = Rear, 0 = Front

Note: Each of the following dummy variables would represent the value 1 against the reference level 0. Details for each of them are included in Appendix

- **Body Style:** Reference = Convertible; Dummies = Hardtop, Hatchback, Sedan, Wagon
- **Drive Wheels:** Reference = 4WD; Dummies = FWD, RWD
- **Engine Type:** Reference = DOHCV; Dummies = L, OHC, OHCF, OHCV, Totor

- **Number of Cylinders:** Reference = Eight; Dummies = Five, Four, Six, Three, Twelve, Two

The results of the model show statistically significant associations with the price of an automobile, indicating that they are meaningful predictors when controlling for other variables (for convenience purposes the influence would be separated between positive and negative effects on the price):

Positive Significant Predictors:

- **Width** (Estimate = 568.6, p = 0.027): A one-inch increase in vehicle width is associated with a average price increase of $568, holding other variables constant.
- **Curb Weight** (Estimate = 3.91, p = 0.033): Each additional pound of vehicle weight is linked to an average price increase of $3.91 price increase, holding other variables constant, suggesting heavier cars tend to be more expensive.
- **Engine Size** (Estimate = 58.05, p = 0.0129): A larger engine size by one cc increases price by $58.05 on average, holding other variables constant.
- **Highway MPG** (Estimate = 318.8, p = 0.0307): Each additional mile per gallon in highway fuel efficiency is associated with an average price increase of $318.
- **Peak RPM** (Estimate = 1.76, p = 0.0044): Holding all other variables constant, for every one-unit increase in peak RPM, the price of a vehicle increases by approximately $1.76 on average. Vehicles that deliver peak power at higher RPM tend to be priced higher.
- **Engine Location Rear** (Estimate = 7521, p = 0.0038): Rear-engine cars, compared to front-engine ones (the reference group), are significantly more expensive. Compared to the reference category (vehicles with front-engine configurations), rear-engine vehicles have a difference of $7,521 in the mean price, when all other predictors are held constant.
- **Engine Type OHC** (Overhead Camshaft) and **Engine Type OHFC** (OverHead Cam and Valve F engine) (Estimates = 2964 and 3712, p = 0.002 and 0.017, respectively): These engine types are positively associated with higher prices relative to the baseline engine type. These positive coefficients indicate that, relative to the baseline engine type, vehicles with OHC or OHCF engines have a difference in the mean price of $2,964 and $3,712, respectively, when all other predictors are held constant.

Negative Significant Predictors:

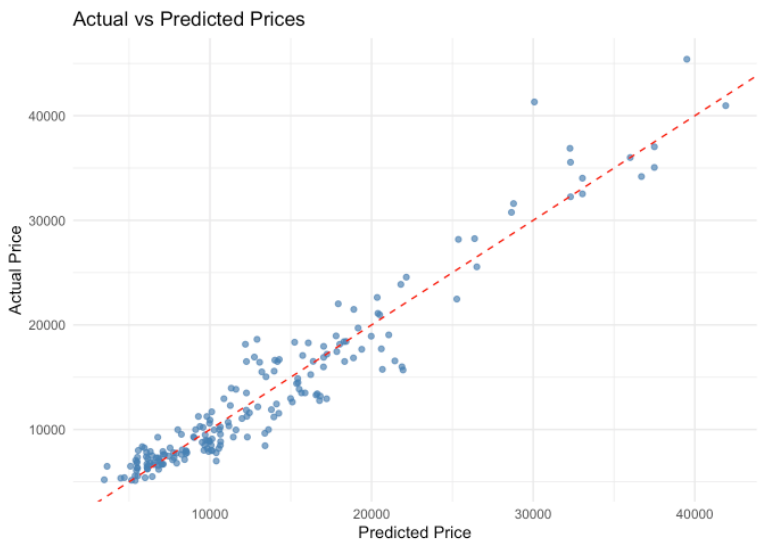
- Several Body Style types such as hardtop, hatchback, sedan, and wagon, have negative and significant coefficients, indicating they are generally associated with lower mean prices than convertibles (reference level), when all other predictors are held constant.
- **Cylinder count** variables such as Number of Cylinders Four, Five and Six also show large negative and highly significant effects on mean price compared to the eight cylinders models, suggesting that fewer cylinders generally correspond with cheaper vehicles, when all other predictors are held constant.
- **Number of Cylinders - Twelve** (Estimate = -8380, p = 0.0145): This is a surprising result because it is counter-intuitive, since it has a negative coefficient but is significant. *Refer to the limitations section for an explanation*.

Results indicate that technical specifications, such as engine size, weight, and dimensions, have measurable and significant effects on vehicle pricing. These relationships align with market intuition where larger, more powerful,

and more efficient vehicles tend to command higher prices. Body style and engine configuration also play key roles, reflecting consumer preferences and manufacturing costs.

Model Assessment

We built a scatter plot to better assess the validity of the model. The "Actual vs Predicted Prices" plot compares the actual car prices in the dataset (y-axis) to the predicted prices generated by the regression model (x-axis). The red dashed line represents the ideal scenario where predicted values perfectly match actual values



Most points fall close to the red line, indicating that the model is performing well and predicting prices accurately for most observations. Note, some of the highest-level cars deviate from the line, meaning that the model might underestimate or overestimate prices for some expensive vehicles. This plot supports the conclusion that the model has a strong overall fit, with most residual errors being relatively small. Residual Standard Error (RSE) was reported as 2410. This means that, on average, the model's predicted prices differ from actual prices by approximately $2,410. Additionally, the coefficient of variation of the model is about 18.25 %, suggesting the on average, the prediction error (standard deviation of residuals) is about 18.25% of the average car price. This indicates a good fit, demonstrating a fair level of precision with some margin of improvement.

The model produced a Multiple R-squared value of 0.9246, indicating that approximately 92.5% of the variability in automobile prices is explained by the model. The Adjusted R-squared, which penalizes for the number of predictors, remains strong at 0.9112, suggesting the model does not suffer significantly from overfitting. However, this also highlights room for improvement. Furthermore, the F-statistic value of 68.94 with 29 and 163 degrees of freedom, plus a p-value < 2.2e-16 provides a strong evidence that the overall regression model is statistically significant, meaning at least one of the predictor variables in the model significantly contributes to explaining the variation in car price, confidently rejecting the null hypothesis.

To further assess model adequacy, we examined several diagnostic plots:

- The Residuals vs. Fitted plot revealed a slight funnel shape, indicating possible heteroscedasticity (non-constant variance). Some residuals appeared notably large, especially for higher fitted values.
- The Q-Q plot demonstrated that residuals mostly follow a normal distribution, with slight deviations at the tails, which is typical for real world data.
- The Scale-Location plot reinforced the potential issue of heteroscedasticity, where variance increases with fitted values.
- Cook's distance and Leverage plots uncovered a few influential observations (notably observations 16, 72, 133, and 160), which could disproportionately affect model estimates.

Based on the previous description, while the model generally fits well, it is worth mentioning that there may be outliers or non-constant variance that could bias or distort the model's predictions.

Limitations

Overall, the model shows a good fit and predictive power. The low RSE and high Adjusted R squared indicate that it is a good model for explaining automobile price variation. However, some limitations need to be addressed, described as following:

Multicollinearity among predictors (engine size and horsepower or wheelbase and length) can inflate standard errors and reduce individual variable effects. This weakens the precision of coefficient estimates, even if overall model fit remains strong.

The exclusion of eight observations due to missing data, given that we started with a relatively considerable sample size, may introduce bias if the missingness is not random, potentially affecting the model's generalizability. Additionally, categorical variables like engine type and number of cylinders rely on reference categories, which must be clearly identified for proper interpretation. The variable "makes" and "Fuel System" were excluded to avoid overfitting due to its many levels.

Finally, diagnostic plots reveal issues such as heteroscedasticity and influential outliers, which may distort results. These factors suggest the need for caution in interpretation and possible refinement through sensitivity testing or alternative modeling techniques.

Summary

Using the dataset from the UCI Machine Learning Repository, we developed and evaluated a multiple linear regression model to identify important factors that influence car prices. Using dummy encoding to include both quantitative and categorical variables, our model was able to include a wide range of vehicle attributes that affect price, such as vehicle configuration, engine specifications, fuel, efficiency and dimensions.

The final model achieved a strong explanatory power, with an adjusted R-squared of 0.9112, indicating over 91% of the variability in car prices is accounted for by the chosen predictors. Variables such as engine size, curb weight, width, highway MPG, and rear engine location appeared as particularly influential in increasing price, while certain body styles and cylinder counts have strong negative association. Diagnostic evaluation supported the model's overall validity while pointing out some potential issues. Plots revealed mild heteroscedasticity, slightly irregular residual pattern, and presence of influential data points that could influence the estimate of the coefficients. They recommend caution when extrapolating the findings, but they do not dispute our model.

While the data is good and helped us to create the model, it also has some limitations that impact inferential robustness. Categorical variables that we removed from the model such as "Make" and "Fuel system" to avoid model overfitting, may omit potentially meaningful predictors. Despite the high overall fit of the model, multicollinearity among correlated predictors such as engine size and horsepower may reduce the precision of individual coefficient estimates.

In conclusion, our model effectively captures the primary drivers of automobile pricing and demonstrates strong predictive performance. Further improvements in data variability, variable selections, and modeling techniques could enhance its practical and inferential accuracy.

References

UCI Machine Learning Repository. *Automobile Data Set.* University of California, Irvine, School of Information and Computer Sciences, 1987. https://archive.ics.uci.edu/dataset/10/automobile.

Appendix

The R code that was used to load, clean, analyze, and model the Automobile Dataset from the UCI Machine Learning Repository will be attached in html file for a better layout. This appendix supports the findings and results discussed in the main body of the report, particularly the modeling and diagnostics section. The code above fully supports the results that are present within the *Model and Interpretation* section of the report, including the dummy variable encoding and correlation analysis. The multiple linear regression model output and diagnostics discussed in the Results and Summary sections are directly produced using the lm() function and residual diagnostic plots.

Correlation Matrix with Respect to Price

To evaluate which quantitative variables may have the strongest linear relationship with car price, we computed correlation coefficients. The table below summarizes the strength and direction associated between numeric predictors and price.

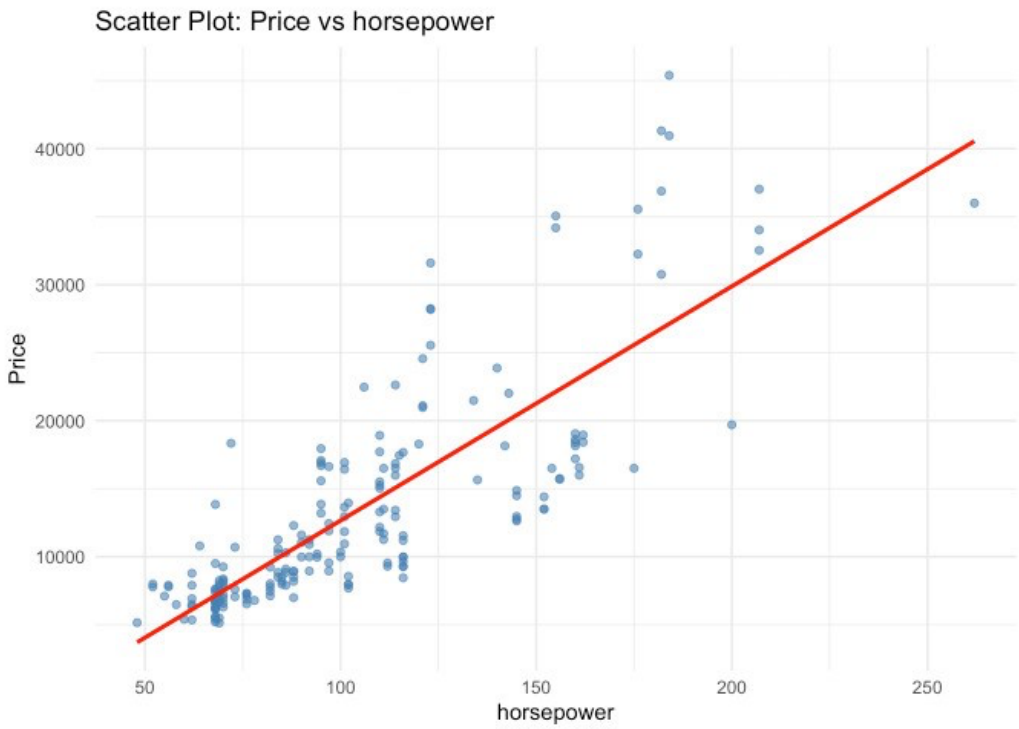
symboling	normalized_losses	wheel_base	length
-0.1633293	0.1999239	0.7347888	0.7603228
width	height	curb_weight	engine_size
0.8433157	0.2475002	0.8938095	0.8417248
bore	stroke	compression_ratio	horsepower
0.5348908	0.1587982	0.2109484	0.7585823
peak_rpm	city_mpg	highway_mpg	
-0.1739701	-0.6900998	-0.7183144	

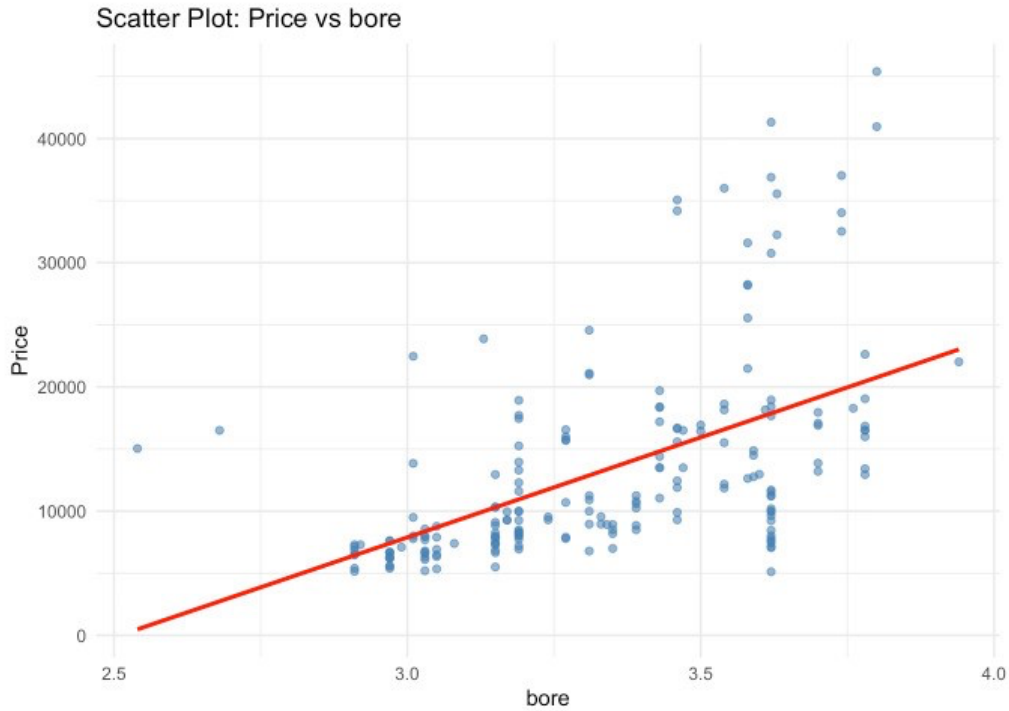
Scatter Plots of Price vs Predictors

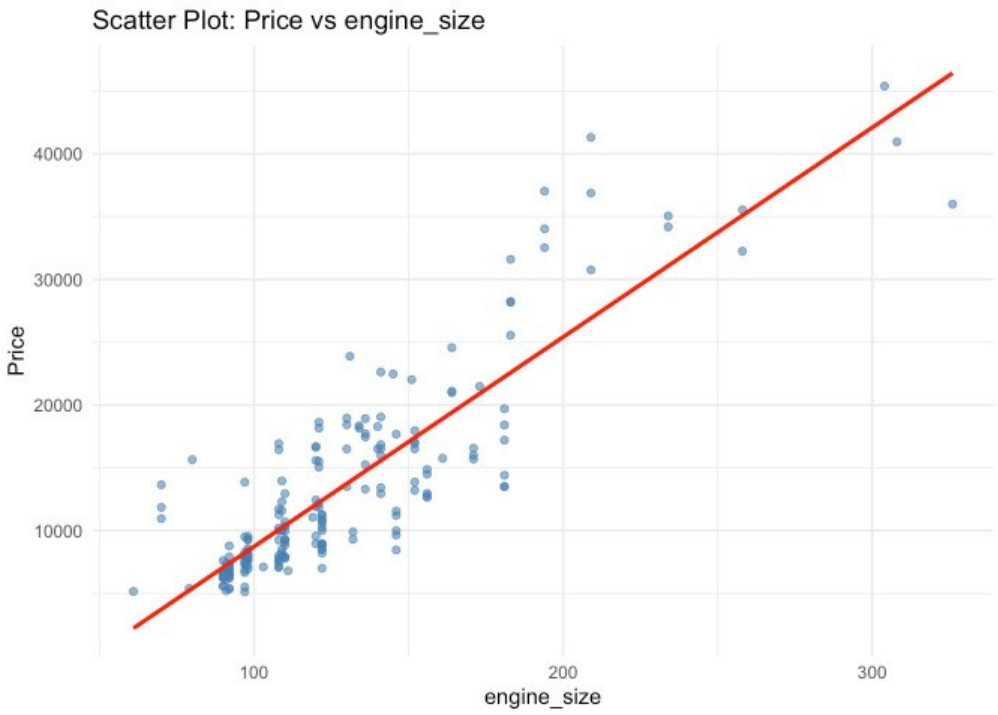
The relationship between each quantitative predictor and car price using scatter plots with fitted regression lines was also used to visualize these relationships. The scatterplots are as follows:

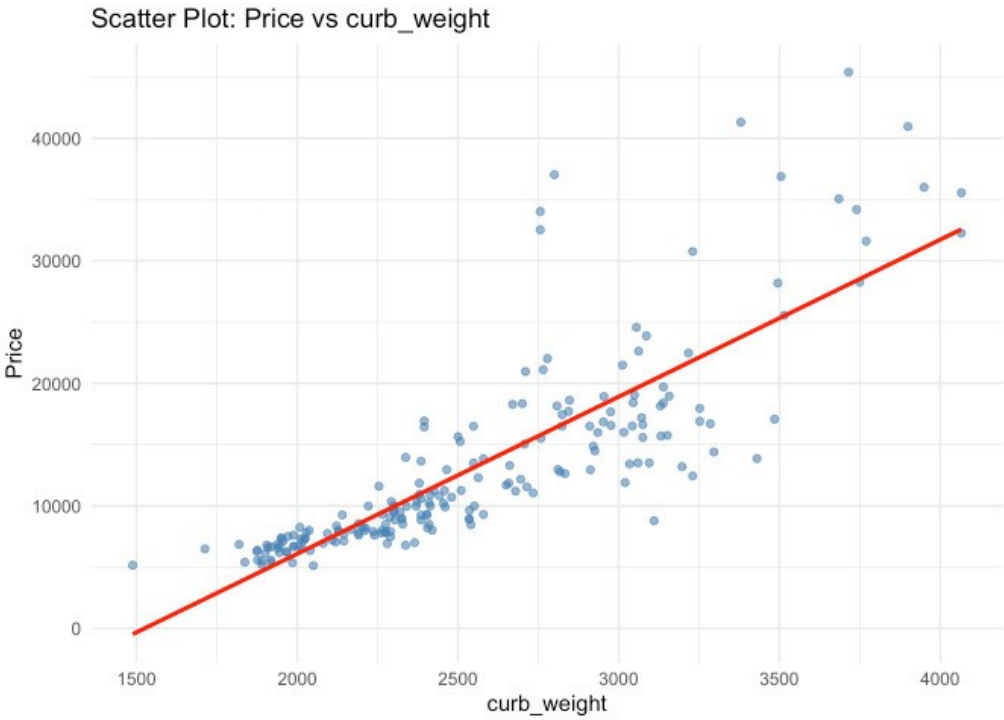
Scatter Plot: Price vs highway_mpg
Price
40000
30000
20000
10000
0
20
30
40
50
highway_mpg

Scatter Plot: Price vs city_mpg
Price
40000
30000
20000
10000
0
20
30
40
50
city_mpg


Scatter Plot: Price vs horsepower
Price
40000
30000
20000
10000
50
100
150
200
250
horsepower

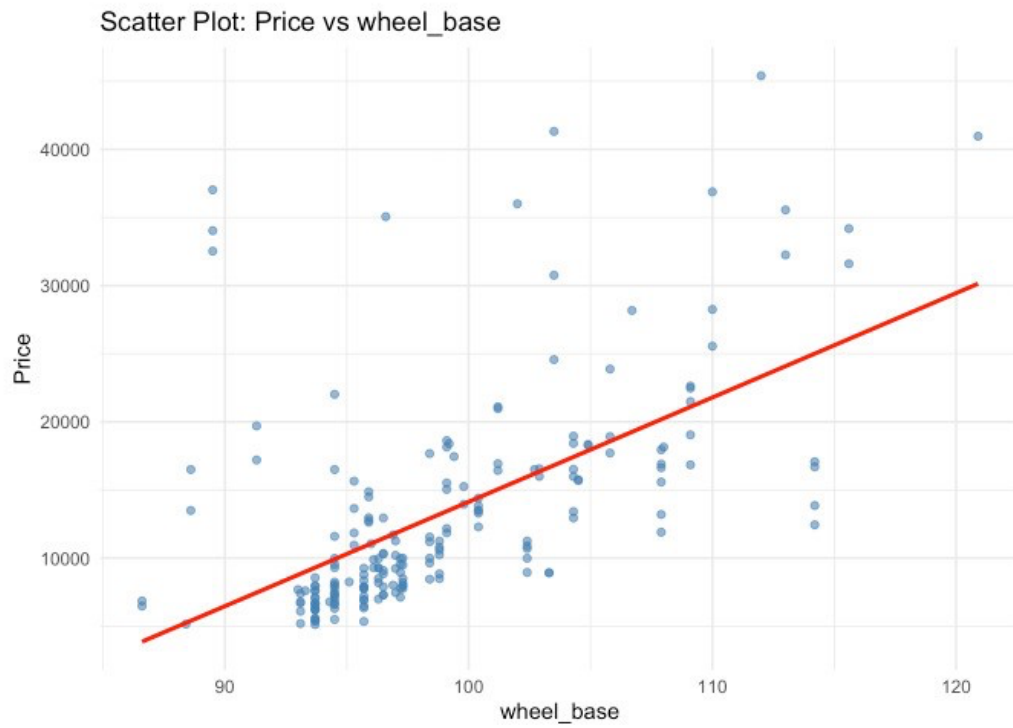

Scatter Plot: Price vs bore
Price
40000
30000
20000
10000
0
2.5
3.0
3.5
4.0
bore


Scatter Plot: Price vs engine_size
Price
40000
30000
20000
10000
100
200
300
engine_size


Scatter Plot: Price vs curb_weight
Price
40000
30000
20000
10000
0
1500
2000
2500
3000
3500
4000
curb_weight

Scatter Plot: Price vs width
Price
40000
30000
20000
10000
0
60.0
62.5
65.0
67.5
70.0
72.5
width

Scatter Plot: Price vs length
Price
40000
30000
20000
10000
0
140
160
180
200
length



Positive linear relationships with price were shown with certain variables such as horsepower, engine size, curb weight, width, length, wheelbase, and bore. Negative linear relationships with price were shown with city MPG and highway MPG. Vehicles with better fuel efficiency tend to be more affordable-likely due to their smaller engine sizes and less premium builds.

Model Structure

The following is the example of the form that this multiple linear model will take:

$$\hat{y} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \varepsilon$$

Where:

- $\hat{y}$ is the predicted automobile price.
- $X_1, X_2, X_3 \ldots\ldots, X_k$ are the predictor variables
- β_0 is the intercept
- $\beta_1 \ldots, \beta_k$ are the coefficients estimated from the data
- E is the error term

1. The relationship between our response (Y) and predictors (X) is linear.
2. The observations are independent given the predictor values.
3. The error values are normally distributed ($\epsilon i \sim$ Normal).
4. Expected value of errors is 0 ($E[\epsilon i] = 0$)
5. Error values have constant standard deviation ($E[\sqrt{(Var(\epsilon i))}] = \sigma$).
6. The error terms are independent.

Model Equation

$$\text{Price} = \beta_0 + \beta_1 * \text{Wheel Base} + \beta_2 * \text{Length} + \beta_3 * \text{Width} + \beta_4 * \text{Curb Weight} + \beta_5 * \text{Engine Size} + \beta_6 * \text{Bore} + \beta_7 * \text{Horsepower} + \beta_8 \cdot \text{City MPG} + \beta_9 * \text{ Highway MPG} + \beta_{10} * \text{Peak RPM } + \sum_j \gamma_j * \text{D}_j + \varepsilon$$

Where:

- D_j are the dummy coded categorical variables
- γ_j are the corresponding coefficients
- E is the error term

Model Equation

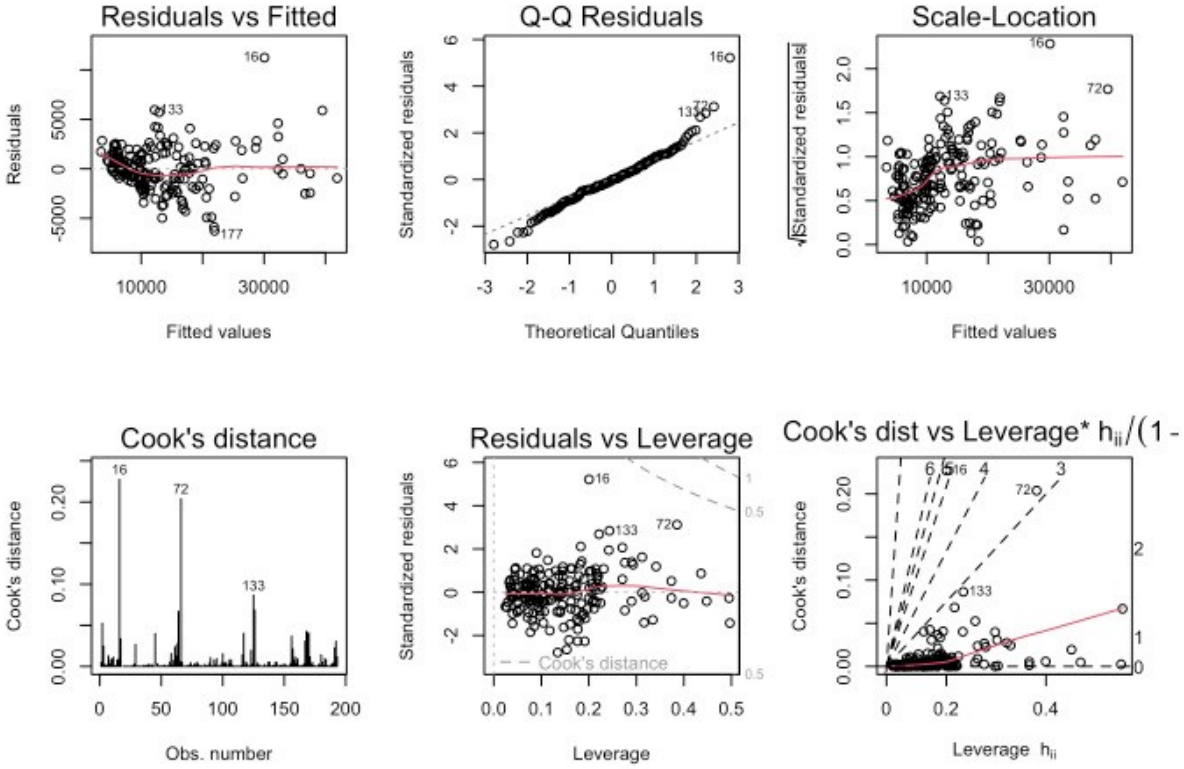
$$\text{Price} = -43780 + 112.1 * \text{wheel}_{\text{base}} - 12.64 * \text{length} + 568.6 * \text{width} + 3.909 * \text{curb}_{\text{weight}} + 58.05 * \text{engine}_{\text{size}} + -1027 * \text{bore} + 21.73 * \text{horsepower} - 282.1 * \text{city}_{\text{mpg}} + 318.8 * \text{highway}_{\text{mpg}} + 1.764 * \text{peak}_{\text{rpm}} + (-1062 * \text{fuel}_{\text{typegas}} + 866.8) * \text{aspirationturbo} + 323.9 \cdot \text{num}_{\text{doorstwo}} + (-4278 * \text{body}_{\text{stylehardtop}} - 4111 * \text{body}_{\text{stylehatchback}} - 3220 * \text{body}_{\text{stylesedan}} - 4277 * + -1243 * \text{drive}_{\text{wheelsfwd}} + 508.4 * \text{drive}_{\text{wheelsrwd}} + 7521 \cdot \text{engine}_{\text{locationrear}} + 126.1 * \text{engine}_{\text{typel}} + 2964 * \text{engine}_{\text{typeohc}} + 3712 \cdot \text{engine}_{\text{typeohcf}} - 4544 * \text{engine}_{\text{typeohcv}} + (-13520 * \text{num}_{\text{cylindersfive}}) - 15840 * \text{num}_{\text{cylindersfou}}\text{r} - 11040 * \text{num}_{\text{cylinderssix}} - 7567 * \text{num}_{\text{cylindersthree}} - 8380 * \text{num}_{\text{cylinderstwelve}}$$

Notes:

- Each binary categorical variable (like fuel_typegas or engine_typeohc) takes a value of 1 if true and 0 otherwise.
- The intercept of -43,780 represents the predicted price when all variables (including all dummies) are at their reference levels or zero.
- Reference levels (automatically assigned by R and omitted from the model – Refer to main body)

Regression Diagnostics

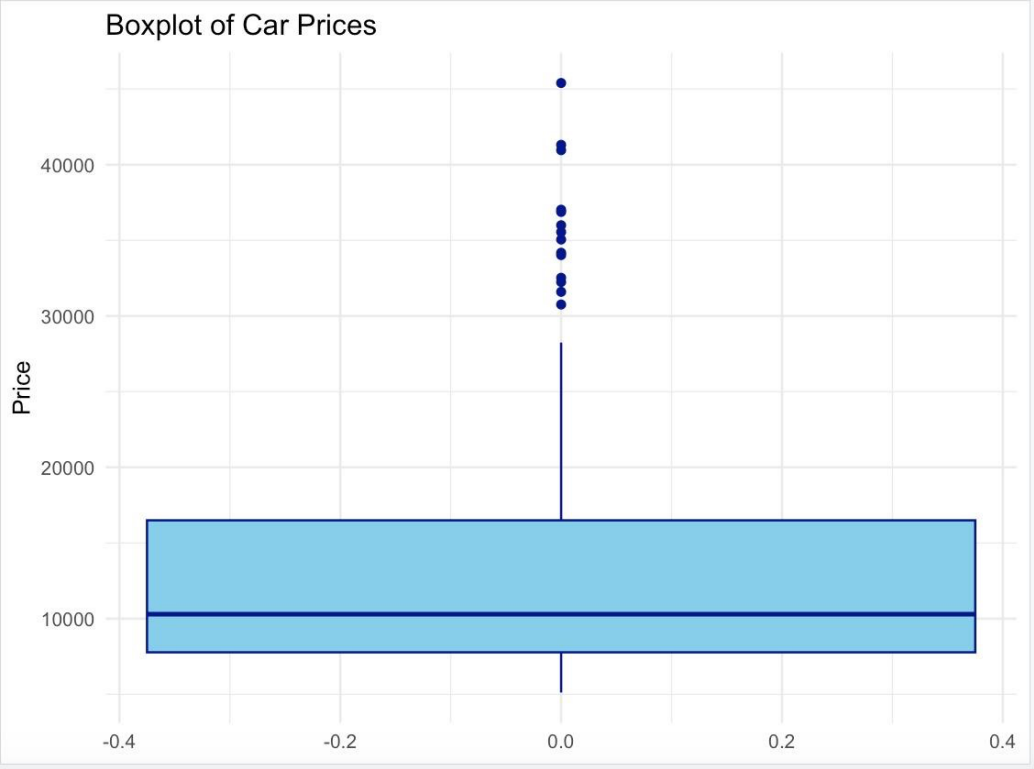
To evaluate the assumptions and reliability of our multiple linear regression model, we generated six standard diagnostic plots. These plots helped assess linearity, normality of residuals, homoscedasticity, and identify influential points. These plots are shown below.

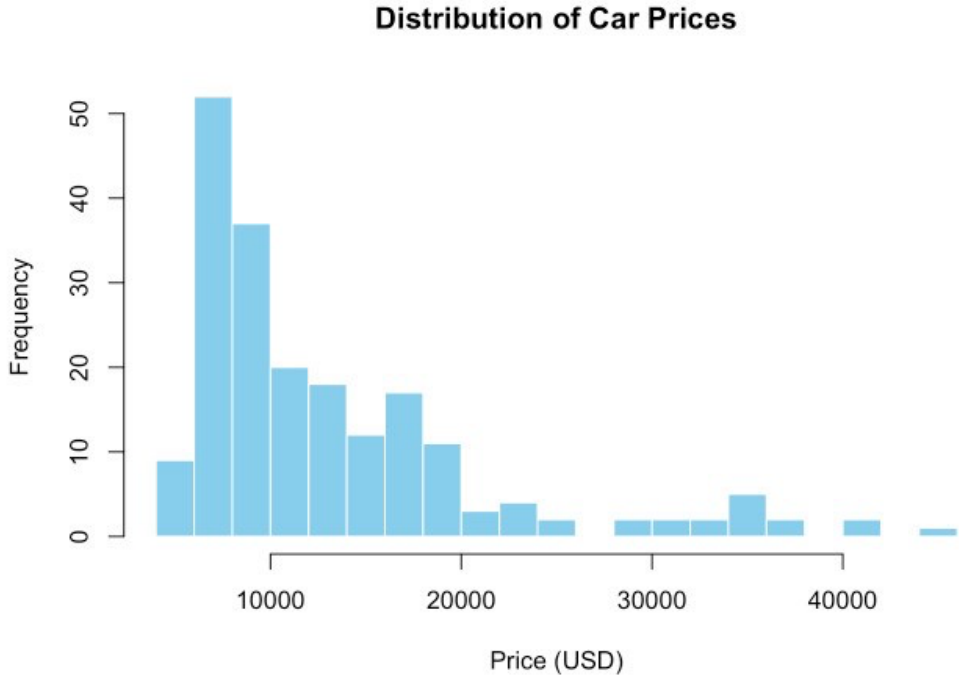


Price Distribution and Outlier Detection

A box plot and scatter plot were created in order to describe data skewness and detect outliers. The box plot highlights a right-skewness distribution in car prices, with several high-end vehicles marked as outliers. The histogram confirms the skewed nature of the price distribution, with most cars' prices under $15,000.

These plots justify the need for caution when interpreting model residuals, especially when assessing fit in the higher price range.


Boxplot of Car Prices
Price
40000
30000
20000
10000
-0.4
-0.2
0.0
0.2
0.4


Distribution of Car Prices
Frequency
0
10
20
30
40
50
10000
20000
30000
40000
Price (USD)

Group Member Contributions Statement with Signatures

The contributions of each group member to the project are summarized as follows:

- Jonida Qira: 25% - contributed to finding datasets, writing the report and editing, and preliminary analysis

- Jordyn Labbe: 25% - contributed to finding datasets, developing model, creating presentation, and writing report

- Uriel Ulloa: 25% - responsible for data collection, developing model, model interpretation, and model validation

- Ryan Le: 25% - contributed to developing model, model validation, datasets finding, and writing report

All group members actively participated in meetings, shared ideas, and reviewed the final document to ensure the quality and completeness of the project.